TUTORIAL – 4

Q1. In DES, given the input of the 8 S-boxes as follows. Determine the output of the 8 S-boxes.

011000 010001 011110 111010 100001 100110 010100 100111

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
S_1																
0	14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
1	0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
2	4	1	14	8	13	6	2	11	15	12	9	7	3	10	5	0
3	15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13
S_2																
	15	1	8	14	6	11	3	4	9	7	2	13	12	0	5	10
	3	13	4	7	15	2	8	14	12	0	1	10	6	9	11	5
	0	14	7	11	10	4	13	1	5	8	12	6	9	3	2	15
	13	8	10	1	3	15	4	2	11	6	7	12	0	5	14	9
S_3																
0	10	0	9	14	6	3	15	5	1	13	12	7	11	4	2	8
1	13	7	0	9	3	4	6	10	2	8	5	14	12	11	15	1
2	13	6	4	9	8	15	3	0	11	1	2	12	5	10	14	7
3	1	10	13	0	6	9	8	7	4	15	14	3	11	5	2	12
S_4																
	7	13	14	3	0	6	9	10	1	2	8	5	11	12	4	15
	13	8	11	5	6	15	0	3	4	7	2	12	1	10	14	9
	10	6	9	0	12	11	7	13	15	1	3	14	5	2	8	4
	3	15	0	6	10	1	13	8	9	4	5	11	12	7	2	14

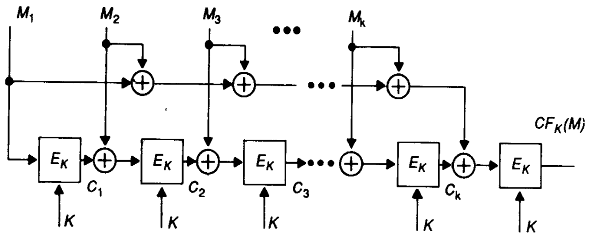
S_5																
	2	12	4	1	7	10	11	6	8	5	3	15	13	0	14	9
	14	11	2	12	4	7	13	1	5	0	15	10	3	9	8	6
	4	2	1	11	10	13	7	8	15	9	12	5	6	3	0	14
	11	8	12	7	1	14	2	13	6	15	0	9	10	4	5	3
S_6																
	12	1	10	15	9	2	6	8	0	13	3	4	14	7	5	11
	10	15	4	2	7	12	9	5	6	1	13	14	0	11	3	8
	9	14	15	5	2	8	12	3	7	0	4	10	1	13	11	6
	4	3	2	12	9	5	15	10	11	14	1	7	6	0	8	13
S_7																
	4	11	2	14	15	0	8	13	3	12	9	7	5	10	6	1
	13	0	11	7	4	9	1	10	14	3	5	12	2	15	8	6
	1	4	11	13	12	3	7	14	10	15	6	8	0	5	9	2
	6	11	13	8	1	4	10	7	9	5	0	15	14	2	3	12
S_8																
	13	2	8	4	6	15	11	1	10	9	3	14	5	0	12	7
	1	15	13	8	10	3	7	4	12	5	6	11	0	14	9	2
	7	11	4	1	9	12	14	2	0	6	10	13	15	3	5	8
	2	1	14	7	4	10	8	13	15	12	9	0	3	5	6	11

Q2. A hash function is shown as follows.

where b_0, b_1, b_2, b_3 are 4 input bits, b_0', b_1', b_2', b_3' are 4 output bits, and K is the constant matrix and vector.

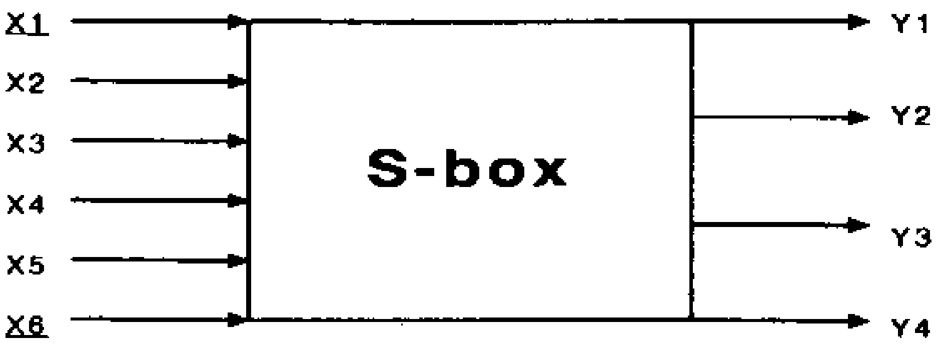
Note that the addition is XOR.

Determine the hash value of (0000, 0001, 0010, 0100).



$$\begin{bmatrix} b_0' \\ b_1' \\ b_2' \\ b_3' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} b_0 \\ b_1 \\ b_2 \\ b_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

E_K is defined as follows.



$(x_1x_2x_3x_4x_5x_6)$==>S-box==>$(y_1y_2y_3\,y_4)$

Row no. = (x_1x_6), Column no. =$(x_2x_3x_4x_5)$

E.g:, In=(110011)→R=(11)=3,C=(1001)=9 Out=4=(0100) y_1=0, y_2=1, y_3=0, y_4=0

S1	S2	S3	S4	S5	S6	S7	S8
011000	010001	011110	111010	100001	100110	010100	100111
R=0 C=12	R=1 C=8	R=0 C=15	R= C=	R= C=	R= C=	R= C=	R= C=
0101	1100	1000	?	?	?	?	?

DE	0	1	2	3	4	5	6	7
BIN	0	1	10	11	100	101	110	111
DE	8	9	10	11	12	13	14	15
BIN	1000	1001	1010	1011	1100	1101	1110	1111

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
								S_1								
0	14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
1	0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
2	4	1	14	8	13	6	2	11	15	12	9	7	3	10	5	0
3	15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13
								S_2								
	15	1	8	14	6	11	3	4	9	7	2	13	12	0	5	10
	3	13	4	7	15	2	8	14	12	0	1	10	6	9	11	5
	0	14	7	11	10	4	13	1	5	8	12	6	9	3	2	15
	13	8	10	1	3	15	4	2	11	6	7	12	0	5	14	9
								S_3								
0	10	0	9	14	6	3	15	5	1	13	12	7	11	4	2	8
1	13	7	0	9	3	4	6	10	2	8	5	14	12	11	15	1
2	13	6	4	9	8	15	3	0	11	1	2	12	5	10	14	7
3	1	10	13	0	6	9	8	7	4	15	14	3	11	5	2	12
								S_4								
	7	13	14	3	0	6	9	10	1	2	8	5	11	12	4	15
	13	8	11	5	6	15	0	3	4	7	2	12	1	10	14	9
	10	6	9	0	12	11	7	13	15	1	3	14	5	2	8	4
	3	15	0	6	10	1	13	8	9	4	5	11	12	7	2	14
								S_5								
	2	12	4	1	7	10	11	6	8	5	3	15	13	0	14	9
	14	11	2	12	4	7	13	1	5	0	15	10	3	9	8	6
	4	2	1	11	10	13	7	8	15	9	12	5	6	3	0	14
	11	8	12	7	1	14	2	13	6	15	0	9	10	4	5	3
								S_6								
	12	1	10	15	9	2	6	8	0	13	3	4	14	7	5	11
	10	15	4	2	7	12	9	5	6	1	13	14	0	11	3	8
	9	14	15	5	2	8	12	3	7	0	4	10	1	13	11	6
	4	3	2	12	9	5	15	10	11	14	1	7	6	0	8	13
								S_7								
	4	11	2	14	15	0	8	13	3	12	9	7	5	10	6	1
	13	0	11	7	4	9	1	10	14	3	5	12	2	15	8	6
	1	4	11	13	12	3	7	14	10	15	6	8	0	5	9	2
	6	11	13	8	1	4	10	7	9	5	0	15	14	2	3	12
								S_8								
	13	2	8	4	6	15	11	1	10	9	3	14	5	0	12	7
	1	15	13	8	10	3	7	4	12	5	6	11	0	14	9	2
	7	11	4	1	9	12	14	2	0	6	10	13	15	3	5	8
	2	1	14	7	4	10	8	13	15	12	9	0	3	5	6	11

Determine the hash value of (0000, 0001, 0010, 0100)

$$\begin{array}{cccc} M1 & M2 & M3 & M4 \end{array}$$

$$C_1 = (E_k \times M_1) \oplus \begin{bmatrix}1\\0\\0\\1\end{bmatrix} = \begin{bmatrix}1&0&1&1\\1&1&0&1\\1&1&1&0\\0&1&1&1\end{bmatrix} \times \begin{bmatrix}0\\0\\0\\0\end{bmatrix} \oplus \begin{bmatrix}1\\0\\0\\1\end{bmatrix} = \begin{bmatrix}0\\0\\0\\0\end{bmatrix} \oplus \begin{bmatrix}1\\0\\0\\1\end{bmatrix} = \begin{bmatrix}1\\0\\0\\1\end{bmatrix}$$

$$C_2 = (E_k \times [M_2 \oplus C_1]) \oplus \begin{bmatrix}1\\0\\0\\1\end{bmatrix} = \begin{bmatrix}1&0&1&1\\1&1&0&1\\1&1&1&0\\0&1&1&1\end{bmatrix} \times \begin{bmatrix}1\\0\\0\\1\end{bmatrix} \oplus \begin{bmatrix}1\\0\\0\\1\end{bmatrix} = \begin{bmatrix}1\\1\\1\\0\end{bmatrix} \oplus \begin{bmatrix}1\\0\\0\\1\end{bmatrix} = \begin{bmatrix}0\\1\\1\\1\end{bmatrix}$$

$$C_3 = (E_k \times [M_3 \oplus C_2]) \oplus \begin{bmatrix}1\\0\\0\\1\end{bmatrix} = \begin{bmatrix}1&0&1&1\\1&1&0&1\\1&1&1&0\\0&1&1&1\end{bmatrix} \times \begin{bmatrix}0\\1\\0\\1\end{bmatrix} \oplus \begin{bmatrix}1\\0\\0\\1\end{bmatrix} = \begin{bmatrix}1\\0\\1\\0\end{bmatrix} \oplus \begin{bmatrix}1\\0\\0\\1\end{bmatrix} = \begin{bmatrix}0\\0\\1\\1\end{bmatrix}$$

$$C_4 = (E_k \times [M_4 \oplus C_3]) \oplus \begin{bmatrix}1\\0\\0\\1\end{bmatrix} = \begin{bmatrix}1&0&1&1\\1&1&0&1\\1&1&1&0\\0&1&1&1\end{bmatrix} \times \begin{bmatrix}0\\1\\1\\1\end{bmatrix} \oplus \begin{bmatrix}1\\0\\0\\1\end{bmatrix} = \begin{bmatrix}0\\0\\0\\1\end{bmatrix} \oplus \begin{bmatrix}1\\0\\0\\1\end{bmatrix} = \begin{bmatrix}1\\0\\0\\0\end{bmatrix}$$

E_K is defined as follows.

$$\begin{bmatrix}b'_0\\b'_1\\b'_2\\b'_3\end{bmatrix} = \begin{bmatrix}1&0&1&1\\1&1&0&1\\1&1&1&0\\0&1&1&1\end{bmatrix} \begin{bmatrix}b_0\\b_1\\b_2\\b_3\end{bmatrix} + \begin{bmatrix}1\\0\\0\\1\end{bmatrix}$$

$$A_{3x2} \cdot B_{2x4} = C_{3x4} = \begin{pmatrix}1&2\\3&4\\5&6\end{pmatrix} \cdot \begin{pmatrix}1&2&3&4\\5&6&7&8\end{pmatrix} = \begin{pmatrix}1\cdot1+2\cdot5 & 1\cdot2+2\cdot6 & 1\cdot3+2\cdot7 & 1\cdot4+2\cdot8\\ 3\cdot1+4\cdot5 & 3\cdot2+4\cdot6 & 3\cdot3+4\cdot7 & 3\cdot4+4\cdot8\\ 5\cdot1+6\cdot5 & 5\cdot2+6\cdot6 & 5\cdot3+6\cdot7 & 5\cdot4+6\cdot8\end{pmatrix} = \begin{pmatrix}11&14&17&20\\23&30&37&44\\35&46&57&68\end{pmatrix}$$

$$\begin{cases} C_1 = E_K(M_1) \\ C_2 = E_K(M_2 \oplus C_1) \\ \quad\vdots \\ C_k = E_K(M_k \oplus C_{k-1}) \\ CF_K(M) = E_K(M_1 \oplus M_2 \oplus \ldots \oplus M_k \oplus C_k) \end{cases}$$

$$CF_{K=4} = (E_k \times [M_1 \oplus M_2 \oplus M_3 \oplus M_4 \oplus C_4]) \oplus \begin{bmatrix}1\\0\\0\\1\end{bmatrix}$$

$$= \begin{bmatrix}1&0&1&1\\1&1&0&1\\1&1&1&0\\0&1&1&1\end{bmatrix} \times \begin{bmatrix}1\\1\\1\\1\end{bmatrix} \oplus \begin{bmatrix}1\\0\\0\\1\end{bmatrix} = \begin{bmatrix}1\\1\\1\\1\end{bmatrix} \oplus \begin{bmatrix}1\\0\\0\\1\end{bmatrix} = \begin{bmatrix}0\\1\\1\\0\end{bmatrix}$$

$$Matrix\ Multiplication\ should\ be\ = 1 * 0\ xor\ 0 * 0\ xor\ 1 * 0\ xor\ 180$$